

A Scaling–Cancellation Principle for Chord-Based Geometric Approximations

C. Wayne Baker
Independent Researcher
Newfoundland and Labrador, Canada

Revised September 11, 2026

Abstract

This paper isolates a fixed-scale error-cancellation mechanism for geometric approximations whose discretization error admits an even-power asymptotic expansion. If

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \cdots,$$

then sampling the same approximation family at scaled resolutions gives predictable factors k^{-2j} on the successive error terms. The two-scale combination

$$\widehat{Q}_{N,k} = \frac{k^2 Q_{kN} - Q_N}{k^2 - 1}$$

therefore cancels the N^{-2} term and satisfies

$$Q - \widehat{Q}_{N,k} = -\frac{C_4}{k^2} N^{-4} + O(N^{-6}).$$

We then formulate the full multi-scale moment system, give its unique Lagrange weights, and derive an explicit leading-error coefficient. For geometric scale nodes $N, bN, \dots, b^s N$, cancellation of the first s even-power terms yields

$$Q - \widehat{Q}_N^{(s)} = (-1)^s C_{2s+2} b^{-s(s+1)} N^{-2s-2} + O(N^{-2s-4}).$$

An equivalent recursive scale ladder gives a geometric analogue of the Richardson–Romberg extrapolation table. Applied to the half-perimeter of a regular inscribed polygon, $Q_N = N \sin(\pi/N)$, the tripling correction gives

$$\pi - \frac{9Q_{3N} - Q_N}{8} = \frac{\pi^5}{1080} N^{-4} + O(N^{-6}).$$

The framework also clarifies the distinction between fixed-resolution scaling–cancellation and nonlinear residual-map refinement. In particular, the proportional-subtended angle-division residual

$$\mathcal{R}_{a,b}(e) = -\frac{a^2 - b^2}{24a^2} e^3 + O(e^5)$$

acts in the signed error variable and multiplies local error order under iteration; it is not an instance of the fixed-scale extrapolation theorem. No claim of new Richardson or Romberg theory is made. The contribution is a geometric formulation, an explicit fixed-scale hierarchy, and a careful separation of the two convergence mechanisms.

Keywords: Richardson extrapolation; geometric approximation; chord approximation; polygonal approximation; error cancellation; N-series; asymptotic expansion.

1 Introduction

Many geometric quantities are approximated by replacing a smooth object with a finite chordal, polygonal, or sampled object. A resolution parameter N then controls the approximation, and in favorable symmetric settings the error has an expansion in even powers of N^{-1} :

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \dots \quad (1)$$

The essential observation is elementary: replacing N by kN multiplies the term N^{-2j} by k^{-2j} . Once those scaling factors are known, linear combinations of several resolutions can annihilate selected terms of the expansion.

This is a geometric use of Richardson extrapolation, not a replacement for it. The point of the present paper is to state the mechanism in a form that is directly reusable across a family of chord-based approximation problems and to separate it cleanly from a second mechanism that occurs in proportional-subtended angle refinement.

The paper makes four concrete steps.

1. It proves the two-scale cancellation formula and records the surviving leading coefficient.
2. It gives the unique multi-scale weights through a Lagrange interpolation formula in the scale variable $x = k^{-2}$.
3. It specializes the construction to geometric nodes $N, bN, \dots, b^s N$, obtaining the explicit factor $b^{-s(s+1)}$ in the first surviving term and an equivalent recursive scale ladder.
4. It compares this fixed-resolution mechanism with a nonlinear cubic angle-residual law, where the active variable is the signed error itself rather than N^{-2} .

The resulting framework is useful for polygonal approximations of π , for chordal perimeter approximations when the required expansion has been proved, and for the geometric N-series constructions developed in companion work. The same algebra also appears in classical extrapolation methods; the present contribution is the geometric organization and the explicit separation of convergence classes.

2 Asymptotic hypothesis and scope

The cancellation statements require an asymptotic expansion. They do not assert that every chord-based approximation possesses one.

Assumption 2.1 (Even-power expansion through order $2r + 2$). *For some integer $r \geq 1$, the approximation family Q_N satisfies*

$$Q - Q_N = \sum_{\ell=1}^{r+1} C_{2\ell} N^{-2\ell} + O(N^{-2r-4}) \quad (N \rightarrow \infty), \quad (2)$$

where the coefficients $C_{2\ell}$ are independent of N .

In particular applications, an expansion of the form (2) may follow from a Taylor expansion, a periodic quadrature argument, Euler–Maclaurin analysis, or a problem-specific exact factorization. Local symmetry often helps produce even powers, but symmetry alone is not taken here as a universal theorem.

Remark 2.2 (Admissible scale factors). Because the index N is typically integral, we use fixed integer scale factors $k \geq 2$ in the basic theorem. More generally, the same algebra works for any fixed scale factors for which all sampled resolutions are admissible.

Remark 2.3 (Claim boundary). The results below are conditional extrapolation statements. They do not supply the expansion (2) for an arbitrary curve, mesh, parameterization, or chord rule. Establishing that expansion remains an application-specific analytic task.

3 The two-scale scaling–cancellation principle

We begin with the elementary scaling law.

Proposition 3.1 (Scaling of the asymptotic coefficients). *Suppose*

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + O(N^{-8}).$$

For a fixed integer $k \geq 2$,

$$Q - Q_{kN} = \frac{C_2}{k^2} N^{-2} + \frac{C_4}{k^4} N^{-4} + \frac{C_6}{k^6} N^{-6} + O(N^{-8}).$$

Proof. Substitute kN for N in the assumed expansion and use the fact that k is fixed. □

Theorem 3.2 (Two-scale scaling–cancellation). *Suppose*

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + O(N^{-6}). \tag{3}$$

For fixed $k \geq 2$, define

$$\widehat{Q}_{N,k} = Q_{kN} + \frac{Q_{kN} - Q_N}{k^2 - 1} = \frac{k^2 Q_{kN} - Q_N}{k^2 - 1}. \tag{4}$$

Then

$$Q - \widehat{Q}_{N,k} = -\frac{C_4}{k^2} N^{-4} + O(N^{-6}). \tag{5}$$

In particular, the order improves from $O(N^{-2})$ to $O(N^{-4})$.

Proof. From (3),

$$Q_N = Q - C_2 N^{-2} - C_4 N^{-4} + O(N^{-6})$$

and

$$Q_{kN} = Q - \frac{C_2}{k^2} N^{-2} - \frac{C_4}{k^4} N^{-4} + O(N^{-6}).$$

Substituting these expressions into (4) gives

$$\widehat{Q}_{N,k} - Q = \frac{C_4}{k^2} N^{-4} + O(N^{-6}),$$

which is equivalent to (5). □

Corollary 3.3 (Tripling). *For $k = 3$,*

$$\widehat{Q}_{N,3} = \frac{9Q_{3N} - Q_N}{8}, \quad Q - \widehat{Q}_{N,3} = -\frac{C_4}{9} N^{-4} + O(N^{-6}). \tag{6}$$

The important point is that C_2 need not be known. Its value disappears because its scale factor k^{-2} is known.

4 Multi-scale cancellation as interpolation in the scale variable

The two-scale formula extends naturally to several resolutions. Let

$$k_0 = 1, \quad k_1, \dots, k_s > 0$$

be distinct admissible scale factors and define

$$x_j = k_j^{-2}.$$

The asymptotic error at resolution $k_j N$ is a power series in x_j :

$$Q - Q_{k_j N} = \sum_{\ell \geq 1} C_{2\ell} x_j^\ell N^{-2\ell}.$$

Thus cancellation of successive even powers is exactly a polynomial interpolation condition at $x = 0$.

Theorem 4.1 (Unique s -term cancellation weights). *Assume*

$$Q - Q_N = \sum_{\ell=1}^{s+1} C_{2\ell} N^{-2\ell} + O(N^{-2s-4}). \quad (7)$$

Let $k_0, \dots, k_s$ be distinct positive admissible scale factors and set $x_j = k_j^{-2}$. Define

$$w_j = \prod_{\substack{m=0 \\ m \neq j}}^s \frac{x_m}{x_m - x_j}, \quad j = 0, \dots, s, \quad (8)$$

and

$$\widehat{Q}_N^{(s)} = \sum_{j=0}^s w_j Q_{k_j N}. \quad (9)$$

Then

$$\sum_{j=0}^s w_j = 1, \quad (10)$$

$$\sum_{j=0}^s w_j k_j^{-2\ell} = 0, \quad \ell = 1, \dots, s, \quad (11)$$

and these conditions determine the weights uniquely. Moreover,

$$Q - \widehat{Q}_N^{(s)} = (-1)^s C_{2s+2} \left(\prod_{j=0}^s k_j^{-2} \right) N^{-2s-2} + O(N^{-2s-4}). \quad (12)$$

Proof. The numbers w_j in (8) are the Lagrange interpolation weights for evaluating at $x = 0$ from the distinct nodes $x_0, \dots, x_s$. Therefore they reproduce constants and annihilate the monomials $x, x^2, \dots, x^s$, which proves (10)–(11). The associated Vandermonde system is nonsingular, so the weights are unique.

Using (7) at each scale and (10)–(11), all terms through N^{-2s} cancel. The first surviving coefficient is

$$C_{2s+2}N^{-2s-2}\sum_{j=0}^s w_j x_j^{s+1}.$$

For $f(x) = x^{s+1}$, the interpolation error identity at $x = 0$ gives

$$0 - \sum_{j=0}^s w_j x_j^{s+1} = \prod_{j=0}^s (0 - x_j) = (-1)^{s+1} \prod_{j=0}^s x_j.$$

Hence

$$\sum_{j=0}^s w_j x_j^{s+1} = (-1)^s \prod_{j=0}^s x_j,$$

which proves (12). $\square$

Remark 4.2. The theorem is a Richardson extrapolation statement written in a geometric scale variable. The explicit product in (12) is useful because it makes the surviving coefficient transparent once the scale nodes are fixed.

5 Geometric scale nodes and the N-series hierarchy

A particularly natural choice in geometric refinement is a fixed base $b \geq 2$ with

$$k_j = b^j, \quad j = 0, 1, \dots, s. \quad (13)$$

These are the resolutions

$$N, bN, b^2N, \dots, b^sN.$$

In the geometric N-series terminology used in companion work, these are the successive scale samples.

Corollary 5.1 (Geometric-node leading coefficient). *Under the hypotheses of Theorem 4.1, choose $k_j = b^j$. Then*

$$Q - \widehat{Q}_N^{(s)} = (-1)^s C_{2s+2} b^{-s(s+1)} N^{-2s-2} + O(N^{-2s-4}). \quad (14)$$

Proof. Since

$$\prod_{j=0}^s k_j^{-2} = \prod_{j=0}^s b^{-2j} = b^{-2\sum_{j=0}^s j} = b^{-s(s+1)},$$

the result follows from (12). $\square$

The same hierarchy can be generated recursively, without solving the full Vandermonde system at every order.

Theorem 5.2 (Recursive fixed-base cancellation ladder). *Let $Q_N^{[0]} = Q_N$. For a fixed integer $b \geq 2$, define recursively*

$$Q_N^{[r]} = \frac{b^{2r} Q_{bN}^{[r-1]} - Q_N^{[r-1]}}{b^{2r} - 1}, \quad r \geq 1. \quad (15)$$

If

$$Q - Q_N = \sum_{\ell=1}^{r+1} C_{2\ell} N^{-2\ell} + O(N^{-2r-4}),$$

then

$$Q - Q_N^{[r]} = O(N^{-2r-2}).$$

The value $Q_N^{[r]}$ is a linear combination of the $r + 1$ geometric-scale samples

$$Q_N, Q_{bN}, \dots, Q_{b^r N}$$

and agrees with the unique estimator of Corollary 5.1.

Proof. Assume inductively that

$$Q - Q_N^{[r-1]} = D_{2r} N^{-2r} + O(N^{-2r-2}).$$

Then

$$Q - Q_{bN}^{[r-1]} = D_{2r} b^{-2r} N^{-2r} + O(N^{-2r-2}).$$

Substitution into (15) cancels the N^{-2r} term, leaving $O(N^{-2r-2})$. The recursion uses exactly the nodes $N, bN, \dots, b^r N$ and satisfies the same moment conditions as the unique Lagrange estimator, so the two forms agree. $\square$

6 Regular polygon approximation of π

For the unit circle, let

$$P_N = N \sin\left(\frac{\pi}{N}\right). \quad (16)$$

This is half the perimeter of the regular inscribed N -gon, so $P_N \rightarrow \pi$. The sine expansion gives

$$\pi - P_N = \frac{\pi^3}{6} N^{-2} - \frac{\pi^5}{120} N^{-4} + \frac{\pi^7}{5040} N^{-6} - \frac{\pi^9}{362880} N^{-8} + O(N^{-10}). \quad (17)$$

Applying Corollary 3.3 gives the one-step tripling estimator

$$\hat{P}_N^{[1]} = \frac{9P_{3N} - P_N}{8} \quad (18)$$

and the explicit error law

$$\pi - \hat{P}_N^{[1]} = \frac{\pi^5}{1080} N^{-4} + O(N^{-6}). \quad (19)$$

Thus the raw N^{-2} polygon error is converted into an N^{-4} error without using the value of the leading coefficient $\pi^3/6$ in the extrapolation formula itself.

At the next level, the three tripling nodes $N, 3N, 9N$ give

$$\hat{P}_N^{[2]} = \frac{P_N - 90P_{3N} + 729P_{9N}}{640}. \quad (20)$$

By Corollary 5.1,

$$\pi - \hat{P}_N^{[2]} = \frac{\pi^7}{3674160} N^{-6} + O(N^{-8}). \quad (21)$$

Equations (19) and (21) are examples of the general 2, 4, 6, $\dots$ order ladder.

Remark 6.1. These formulas do not define a new value of π . They accelerate a known convergent polygonal approximation by cancelling known asymptotic modes.

7 Ellipse and smooth-curve applications

The abstract theorem applies to any chordal perimeter family after an expansion of the required form has been established for that family. One important companion example is the equal-parameter inscribed polygon of a nondegenerate ellipse. In that setting, the companion ellipse analysis gives an exact factorization of the form

$$P_N = \operatorname{sinc}\left(\frac{\pi}{N}\right) M_N, \quad \operatorname{sinc} x = \frac{\sin x}{x}, \quad (22)$$

where M_N is a periodic midpoint approximation to the ellipse speed. For a fixed nondegenerate ellipse, the midpoint remainder is exponentially small, so the algebraic part of the polygon error is governed by the sinc series. Consequently,

$$\frac{P - P_N}{P} = \frac{\pi^2}{6N^2} - \frac{\pi^4}{120N^4} + O(N^{-6}) + O(e^{-\rho N}) \quad (23)$$

for any admissible strip parameter $\rho > 0$ below the analytic width. The tripling correction therefore gives

$$P - \frac{9P_{3N} - P_N}{8} = \frac{P\pi^4}{1080}N^{-4} + O(N^{-6}) + O(e^{-\rho N}). \quad (24)$$

For a more general smooth closed curve, one should not assume (23). Instead, if a particular sampling and chord rule is proved to satisfy

$$P - P_N = C_2N^{-2} + C_4N^{-4} + O(N^{-6}),$$

then Theorem 3.2 applies immediately. This conditional formulation keeps the extrapolation theorem separate from the analytic work needed to justify the expansion in each geometry.

8 A distinct mechanism: nonlinear cubic residual refinement

The fixed-scale cancellation above acts in the resolution variable N^{-2} . A different error-suppression mechanism appears in the proportional-subtended angle-division construction.

For positive integers a and b , define the local transfer map

$$\Phi_{a,b}(x) = 2 \arcsin\left(\frac{a}{b} \sin \frac{x}{2}\right).$$

If e is a signed angular error, the distinguished residual correction is

$$\mathcal{R}_{a,b}(e) = e - \Phi_{a,b}\left(\frac{b}{a}e\right). \quad (25)$$

Its local expansion is

$$\mathcal{R}_{a,b}(e) = -\frac{a^2 - b^2}{24a^2}e^3 + O(e^5), \quad a \neq b. \quad (26)$$

For the motivating 4:3 branch,

$$\mathcal{R}_{4,3}(e) = -\frac{7}{384}e^3 + O(e^5). \quad (27)$$

The distinction from scaling-cancellation is structural. If $e = O(\varepsilon^p)$ and the cubic coefficient in (26) is nonzero, then

$$\mathcal{R}_{a,b}(e) = O(\varepsilon^{3p}).$$

Thus repeated residual correction multiplies the local error order by three:

$$3 \longrightarrow 9 \longrightarrow 27 \longrightarrow 81 \longrightarrow \cdots .$$

By contrast, the fixed-scale hierarchy cancels successive asymptotic powers of N^{-2} :

$$2 \longrightarrow 4 \longrightarrow 6 \longrightarrow 8 \longrightarrow \cdots .$$

Table 1: Two distinct geometric error-suppression mechanisms.

Feature	Scaling–cancellation	Cubic residual refinement
Active variable	N^{-2}	signed error e
Operation	linear combination of scales	nonlinear residual map
Local law	cancel selected powers	$e \mapsto ce^3 + O(e^5)$
Order behavior	$2, 4, 6, 8, \dots$	$3, 9, 27, 81, \dots$
Classical analogue	Richardson–Romberg	nonlinear local iteration

The two mechanisms can coexist within a broader geometric approximation program, but neither should be identified with the other.

9 Classical extrapolation and claim boundary

The algebra of Sections 3–5 belongs to the classical extrapolation family. Richardson’s work established the general principle of using asymptotic error structure to improve approximations, and Romberg later organized repeated extrapolation on geometrically refined quadrature meshes. The recursion (15) is therefore not claimed as a new numerical-analysis principle.

What is specific here is the geometric packaging:

- the resolution variable is attached directly to chordal and polygonal geometric approximants;
- the scale variable $x = k^{-2}$ makes the moment structure explicit;
- geometric nodes yield the closed surviving factor $b^{-s(s+1)}$;
- the framework is placed beside, and explicitly separated from, the nonlinear proportional-subtended cubic residual cascade.

Likewise, the phrase *geometric N -series* is used here as terminology for the author’s scale-indexed geometric approximation framework. It is not a claim that the finite extrapolation formulas are outside classical Richardson-type theory, nor is the fixed-scale mechanism identified with Landen or modular transformations.

10 Conclusion

An even-power geometric error expansion provides a simple scale algebra. If

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + \cdots ,$$

then the samples Q_{kN} carry the same coefficients multiplied by the known factors k^{-2j} . The two-scale estimator

$$\hat{Q}_{N,k} = \frac{k^2 Q_{kN} - Q_N}{k^2 - 1}$$

cancels the leading term. With $s + 1$ distinct scales, Lagrange weights in $x = k^{-2}$ cancel the first s terms, and geometric scale nodes $N, bN, \dots, b^s N$ yield

$$Q - \hat{Q}_N^{(s)} = (-1)^s C_{2s+2} b^{-s(s+1)} N^{-2s-2} + O(N^{-2s-4}).$$

For the regular polygon half-perimeter this gives a direct hierarchy of accelerated approximations to π ; for ellipse and other chordal perimeter problems it applies once the needed expansion has been established. The same framework also provides a clean taxonomic boundary: fixed-resolution Richardson-type cancellation raises the order along an even-power ladder, whereas proportional-subtended residual refinement acts nonlinearly on the signed error and multiplies its local order.

Future work may study stability of the multi-scale weights, optimal scale selection under finite precision or geometric construction cost, and extensions to curved-surface approximation families for which an even-power expansion can be proved.

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